

LAB 1 — Wireless Security Basics (Week 1)

Duration: 60–90 minutes

Purpose

You will observe how Wi-Fi networks advertise themselves and how devices join/roam, then connect what you saw to real threats and realistic defenses. This is a defensive lab (observe + harden), not an offensive lab.

Rules

Only work on the instructor lab SSID(s) or your own hotspot. Do not capture traffic from networks you don't own or have explicit permission to analyze.

What you need

Wireshark installed.

Either:

- (A) Live capture on a permitted Wi-Fi network (lab SSID / your hotspot), or
- (B) Instructor-provided PCAP file(s) if your device/OS can't capture full 802.11 management frames.

Important note about Live Capture (read this once)

On many Windows/macOS setups, Wireshark cannot capture full 802.11 management frames (beacons/probes) unless you have special drivers/monitor mode support. If your live capture does not show what you need, switch to PCAP mode immediately.

Part 1 — Warm-up (10 minutes)

Write short answers (1–2 sentences each).

- 1) Why is wireless generally easier to eavesdrop on than wired?
- 2) Explain in plain words:
eavesdropping, altering/inserting messages, disruption.
- 3) Define (no textbook wording): SSID, access point, station, BSS, ESS.
- 4) In one sentence: what does “association” enable in Wi-Fi?

Part 2 — Wireshark “How-to” Guide (use this while working)

This section tells you exactly what to click and what to type so you can find the required frames.

Step 2.1 — Start Wireshark (choose ONE mode)

PCAP mode (recommended if available):

Open Wireshark → File → Open → choose the .pcapng file.

Live mode:

Open Wireshark → on the start screen choose the interface labeled Wi-Fi/WLAN → click the blue shark fin to start capturing.

If you chose live mode and you see almost only TCP/UDP/IP and nothing that looks like 802.11/wlan, stop and use PCAP mode. That’s normal on many laptops.

Step 2.2 — Use display filters (copy/paste)

Wireshark has “Display Filter” at the top. Paste one filter at a time and press Enter.

Beacon frames (AP “advertisements”): `wlan.fc.type_subtype == 0x08`

Probe requests (device searching): `wlan.fc.type_subtype == 0x04`

Probe responses (AP replies): `wlan.fc.type_subtype == 0x05`

Association request (join request): `wlan.fc.type_subtype == 0x00`

Association response (join answer): `wlan.fc.type_subtype == 0x01`

Reassociation request (roaming): `wlan.fc.type_subtype == 0x02`

802.1X/EAPOL (enterprise auth traffic): `eapol`

If a filter returns 0 packets, try removing it and using the simpler filter:

`wlan`

If “wlan” shows nothing, you are not seeing 802.11 frames in this capture (switch to PCAP).

Step 2.3 — How to “read” a packet (where to find SSID/security)

When you click a packet, look at the middle pane (Packet Details). Expand:
IEEE 802.11 wireless LAN management frame

For beacons and probe responses, find:

- SSID: under “Tagged parameters” (sometimes called “SSID parameter set”)
- BSSID (AP MAC): usually visible in the Address fields (Address 3 or “BSSID” field)
- Security:

Look for RSN information (this indicates WPA2/WPA3 capabilities).

If RSN is absent, the network may be open or using older mechanisms (depends on capture).

Part 3 — Observation Tasks (25–30 minutes)

You will find specific frame types and write down what they mean.

Task A: Beacons (how networks announce themselves)

Find three beacon frames using the beacon filter.

For each beacon, record:

- SSID
- BSSID (AP MAC)
- Channel/frequency (if shown)
- Security capability (open vs WPA2/WPA3 indicated)

Evidence requirement:

Take one screenshot showing a beacon frame where SSID and the security-related fields (e.g., RSN) are visible in the details pane.

Task B: Probes (how devices search for networks)

Find at least one probe request and one probe response.

Answer:

- 1) What is a probe request in simple terms?
- 2) Why can probe requests leak privacy for mobile devices?

Task C: Association (how a device joins)

Find an association request/response OR a reassociation (roaming) if available.

Answer:

- 1) Who is joining whom (station ↔ AP)?
- 2) Why does association/reassociation matter for delivery inside an ESS/DS?

Evidence (if you don't have enterprise PCAP):

If you cannot do Part 5 (EAPOL), take your second screenshot here (association or reassociation frame).

Part 4 — “Signal hiding” vs real protection (15 minutes)

Write one short paragraph (6–10 lines) explaining:

- Why disabling SSID broadcast or using a “weird SSID name” is not real security (it may reduce visibility but doesn't prevent interception).
- Why encryption and proper authentication are real controls.

Mention at least one of these from the lecture: encryption, 802.1X, AP placement/power.

Part 5 — Access control: PSK vs 802.1X (15–25 minutes)

This section is conceptual, plus optional capture proof if you have the enterprise PCAP.

Write full-sentence answers:

- 1) In WPA2-Personal (PSK), who must know the password and why is that risky?
- 2) What happens when a student/employee leaves the organization under PSK?
- 3) What does 802.1X add conceptually compared to PSK (focus on identity/accountability)?
- 4) Why does per-user identity help incident response?
- 5) Give one place where PSK is acceptable and one where 802.1X is preferred.

Optional (only if enterprise_8021x.pcapng is provided):

Open it, apply filter “eapol”, click an EAPOL frame, and take a screenshot.

Evidence requirement for this optional path:

Screenshot showing an EAPOL frame in the packet list and its details pane.

Part 6 — Mobile device security mini-case (10 minutes)

Scenario: A phone connects to public café Wi-Fi, opens a corporate app, then roams to another AP with the same SSID.

Write a short paragraph:

- Name two relevant threats (from eavesdropping / altering/inserting / disruption).
- Propose two defenses: one technical and one policy control.
- Explain why the old “inside safe/outside unsafe” perimeter model breaks down here.

Submission (single file)

Submit one PDF/doc containing:

- Your answers for Parts 1–6.
- Screenshot #1: beacon frame (SSID + security fields visible).
- Screenshot #2: either EAPOL (if you have it) OR association/reassociation.

Troubleshooting (quick fixes students can try)

If you see 0 packets with the beacon/probe filters:

- 1) Clear the filter and try “wlan”.
- 2) If “wlan” still shows nothing, you are not capturing 802.11 frames (switch to PCAP).

If you are in live mode and only see IP/TCP/UDP:

This is a limitation of many Wi-Fi drivers/OS settings. Use the PCAP provided by the instructor.

If your capture is “too big”:

Stop capture after 30–60 seconds, then analyze.

Cheat sheet (filters)

Beacon: wlan.fc.type_subtype == 0x08

Probe req: wlan.fc.type_subtype == 0x04

Probe resp: wlan.fc.type_subtype == 0x05

Assoc req: wlan.fc.type_subtype == 0x00

Assoc resp: wlan.fc.type_subtype == 0x01

Reassoc: wlan.fc.type_subtype == 0x02

Enterprise auth: eapol

Windows note (important):

On many Windows laptops, Wireshark live capture does NOT show full 802.11 management frames (beacons/probes)

because of Wi-Fi driver limitations. If your beacon/probe filters return 0 packets and you mainly see TCP/UDP/IP,

stop live capture and use the provided PCAP file instead. This is expected and not your fault.